SCHOOL OF COMPUTING

(HIGHER NATIONAL DIPLOMA IN SOFTWARE ENGINEERING 25.2F)

FIRST PROGRESS REPORT

Smart Mobile Trash Bin with Dual Navigation Modes (Line Following + Random Obstacle Avoidance)

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1. Problem Identification

In large canteen areas, many people keep eating and moving quickly. When trash bins are placed only in a few fixed locations, some people do not walk to the bin and waste ends up on tables or the floor. This increases cleaning time and creates an unhealthy environment.

This project proposes a mobile trash bin robot that moves around the canteen to reduce the "distance problem". The bin can travel continuously and stop when a user is nearby, allowing waste to be disposed easily without searching for a bin.

1.1 Aim

To design and build a smart mobile trash bin that supports two navigation modes (Line Following and Random Obstacle Avoidance) and provides an automatic lid opening system for user convenience.

1.2 Objectives

- Implement dual navigation modes and a physical switch to select the mode.
- Detect users in front of the bin and open the lid automatically.
- Detect trash level and warn when the bin is full.
- Show robot status (mode, bin level, battery) on a display and with LEDs.
- Support a route loop and automatic return to a docking/charging point (line-based return).

1.3 Scope

The robot is intended for indoor environments (canteen, corridors). The line-following mode uses a black tape line path. Random mode is used for free roaming while avoiding obstacles. The return-to-charge behavior is planned as "return to docking point on the line".

2. Detailed Design (Sketch / 3D Concept)

The robot is a small, wheeled platform carrying a trash container on top. It uses two DC geared motors for movement and a caster wheel for balance. A servo motor opens and closes the lid. Sensors at the bottom read the line, while a front ultrasonic sensor detects obstacles and user hands.

2.1 Main Features

- Dual Mode Navigation: Line Following and Random Obstacle Avoidance.
- Mode Switch: physical toggle/slide switch selects the navigation mode.
- Automatic Lid: stop and open lid when a user hand is detected in front.
- Trash Full Detection: ultrasonic sensor inside the bin measures fill level.
- LED Indicators + Display shows status, mode, battery and bin level.
- Smart Route Loop: continuous loop in line mode for predictable coverage.
- Auto Return to Dock: when battery is low or bin is full, robot returns via line.

2.2 3D Concept Image

A simple concept sketch of the overall physical design is shown below (bin, lid, LCD, sensors and wheelbase).

3D Concept Design (Isometric Sketch

2.3 High-Level Working Concept

The robot continuously moves based on the selected mode. In line mode it follows the floor line. In random mode it moves forward and uses the front ultrasonic sensor to avoid obstacles. At any time, if the robot detects a user hand close to the front sensor, it stops, opens the lid for a few seconds, then closes and continues.

3. Bill of Materials (BOM)

The BOM below lists the main components, quantities and estimated unit costs (LKR).

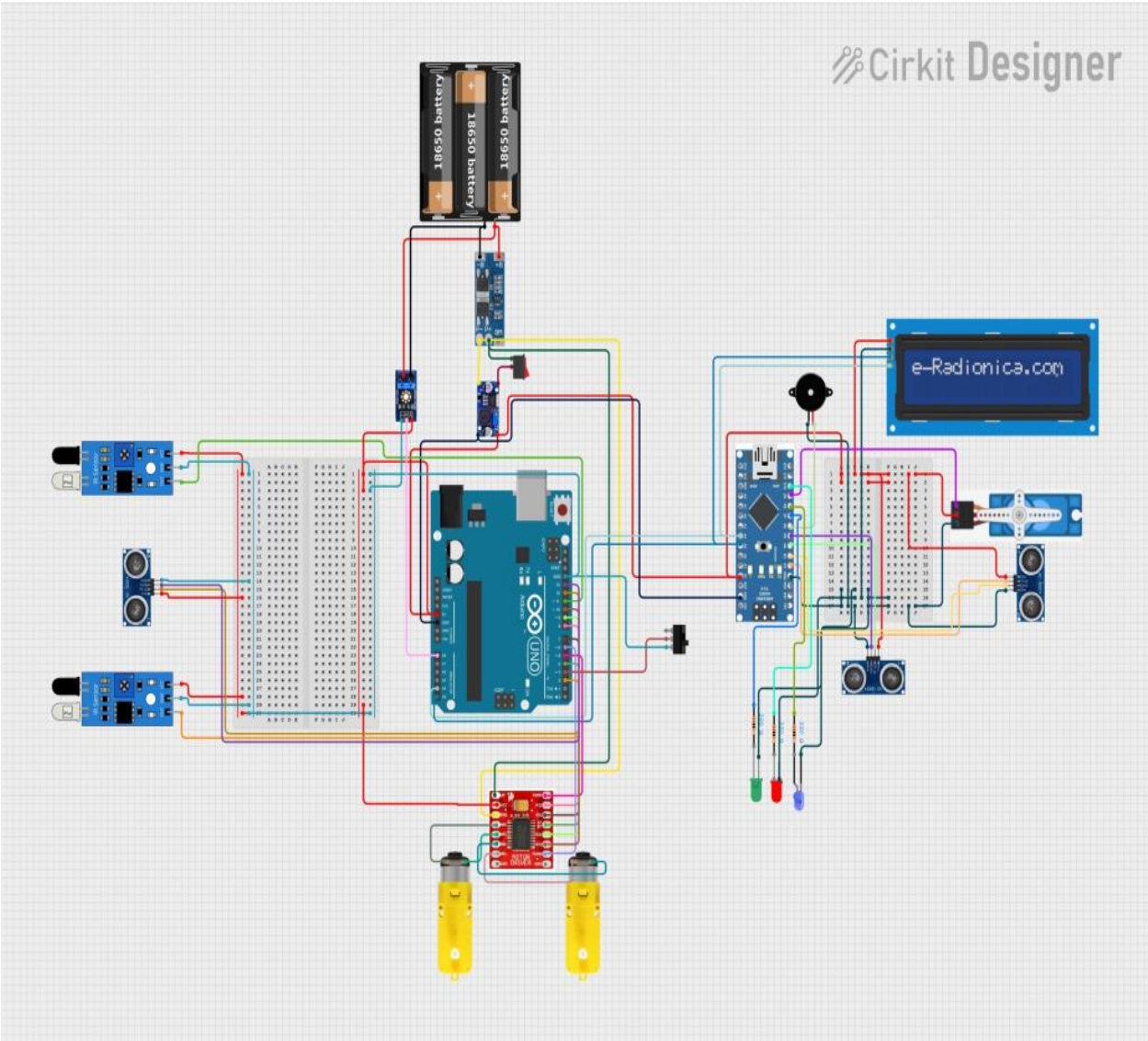
Item ID	Component	Category	Spec / Example	Qty	Unit Price (LKR) (est.)	Total (LKR)	Notes / Where used
C1	Arduino Nano (recommended)	Controller	ATmega328P Nano V3 CH340	1	950	950	Main microcontroller
C3	Motor driver TB6612FNG (recommended)	Motor Driver	Dual DC motor driver module	1	500	500	Efficient driver for 2 DC motors
M1	DC geared motor	Motors	150–300 RPM, 6–12V	2	900	1,800	Left + Right wheels
M2	Wheels	Mechanics	65–80mm diameter (match motors)	2	250	500	Drive wheels
M3	Caster wheel	Mechanics	1 free wheel	1	250	250	Balance support
M4	Chassis / base plate	Mechanics	Acrylic/wood 20–25cm approx.	1	1,200	1,200	Robot base
M5	Motor brackets + screws set	Mechanics	Metal brackets + screws	1	300	300	Mount motors firmly
S1	IR line sensor module (1-channel)	Sensors	TCRT5000 3-sensor line tracker	2	650	1,300	Line following mode
S2	Ultrasonic sensor (front) HC-SR04	Sensors	Obstacle detection	1	240	240	Obstacle avoid
S3	Ultrasonic sensor (inside bin) HC-SR04	Sensors	Mounted inside, facing down	1	240	240	Trash level (bin full)
S4	Ultrasonic sensors (bin front) HC-SR04	Sensors	bin front mounted	1	240	240	hand detect
A1	Servo motor SG90 / MG90S	Actuator	SG90 for light lid (180°)	1	325	325	Open/close lid
A2	Hinge / linkage parts	Actuator	Small hinge + wire/rod	1	200	200	Connect servo to lid
I1	2-position toggle/slide switch	Interface	Mode select: LINE / RANDOM	1	150	150	Connect to Arduino digital pin
I2	LEDs (Green, Blue, Red)	Indicators	5mm LEDs	3	30	90	Status indication
I3	Resistors for LEDs	Indicators	220Ω	3	10	30	Protect LEDs
I4	Buzzer	Indicators	5V active buzzer	1	120	120	Sound feedback
D2	16x2 I2C LCD	Display	LCD with I2C backpack	1	850	850	Alternative to OLED
P1	18650 Li-ion cells	Power	3.7V 2200–3000mAh	2	650	1,300	Battery pack (2S)
P2	18650 holder (2-cell)	Power	2S holder	1	250	250	Holds cells
P3	Battery protection (2S BMS)	Power	2S BMS board	1	350	350	Protection for Li-ion pack
P4	5V buck converter (step-down)	Power	7–12V to stable 5V	1	350	350	Power Arduino + sensors + servo
P5	Main power ON/OFF switch	Power	Rocker switch	1	150	150	Master power
P6	Voltage sensor module / divider	Power	Battery monitoring	1	180	180	Low battery detection
B1	Jumper wires set	Build	Male-Male + Male-Female	1	400	400	Prototyping + wiring
B2	Breadboard (for prototype)	Build	Half-size	1	250	250	Testing stage
B4	Heat shrink / insulation tape	Build	Wire protection	1	200	200	Safety
B5	Cable ties / zip ties	Build	Wire management	1	150	150	Clean build
B6	Hot glue / double tape	Build	Mount sensors + parts	1	200	200	Assembly
T1	Plastic bin container	Body	Small box/bin	1	300	300	Trash bin body
T2	Bin inner ultrasonic mount bracket	Body	Acrylic/3D print (optional)	1	150	150	Optional mount
P1	Board Module LCD Display (PCF8574T)	Display	Interface Board Module LCD Display	1	150	150	converts I2C serial data to parallel data for the LCD display.

Totals	
Total (All items)	13,665

Note: Full BOM with all items and totals is available in the attached spreadsheet. A screenshot of the full table is also included below for quick viewing.

4. Circuit Diagram (Simulation)

The circuit was prepared using a simulation-style wiring tool. The design uses Arduino UNO as the main controller, TB6612FNG motor driver for two DC motors, a 2S battery pack with 2S BMS protection, and a 5V buck converter to supply stable 5V to the Arduino, sensors, servo and display.



4.1 Key Pin Connections (Summary)

Motor Driver (TB6612FNG) control pins (example mapping):

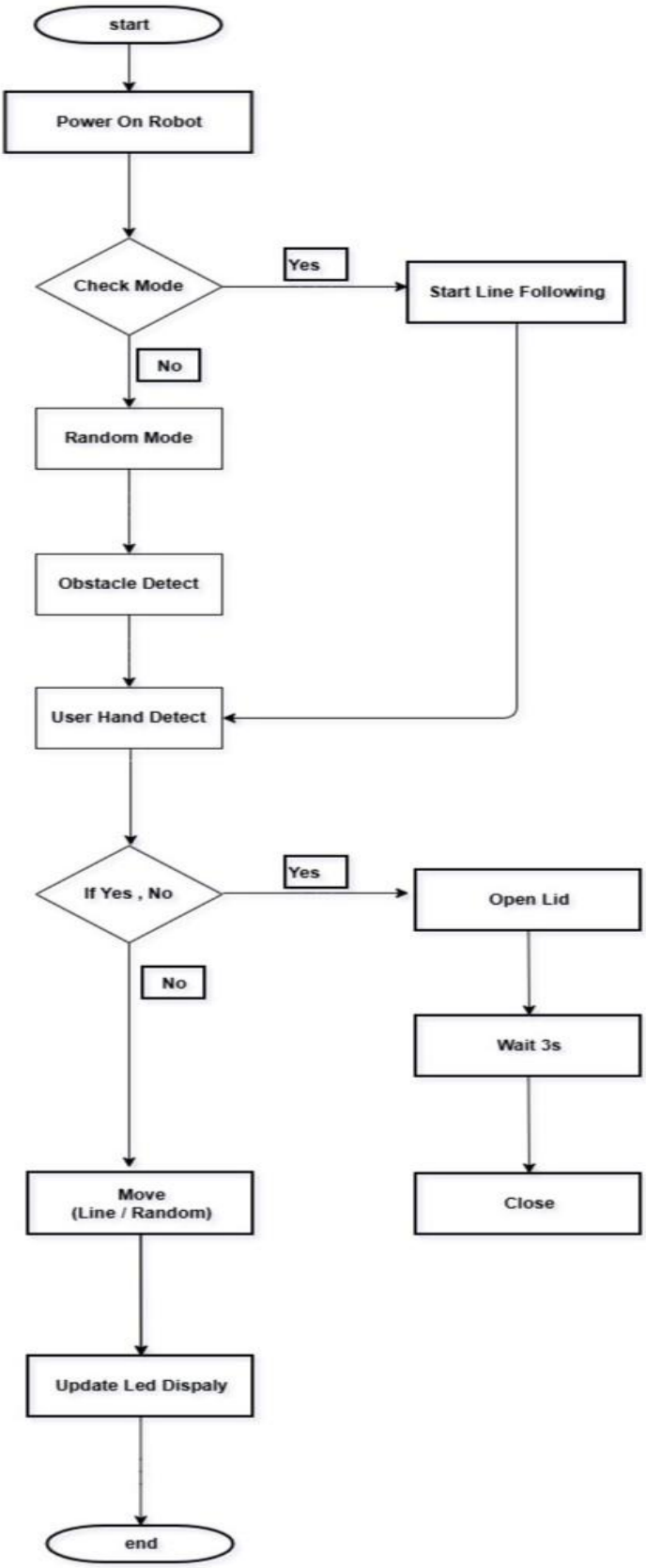
- PWMA → D5, AIN1 → D4, AIN2 → D7 (Left motor)
- PWMB → D6, BIN1 → D8, BIN2 → D12 (Right motor)
- STBY → 5V (enabled), VCC → 5V (logic), VM → Battery (motor power), GND → Common GND

Sensors and UI (example mapping):

- Line sensors (2 units): Left OUT → A0, Right OUT → A1
- Front Ultrasonic (Obstacle/User): TRIG → D10, ECHO → D11
- Bin Ultrasonic (Full detection): TRIG → D3, ECHO → A3
- Servo (Lid): Signal → D9
- Mode Switch: D13 with INPUT_PULLUP (switch to GND = LINE mode)
- LCD I2C: SDA → A4, SCL → A5

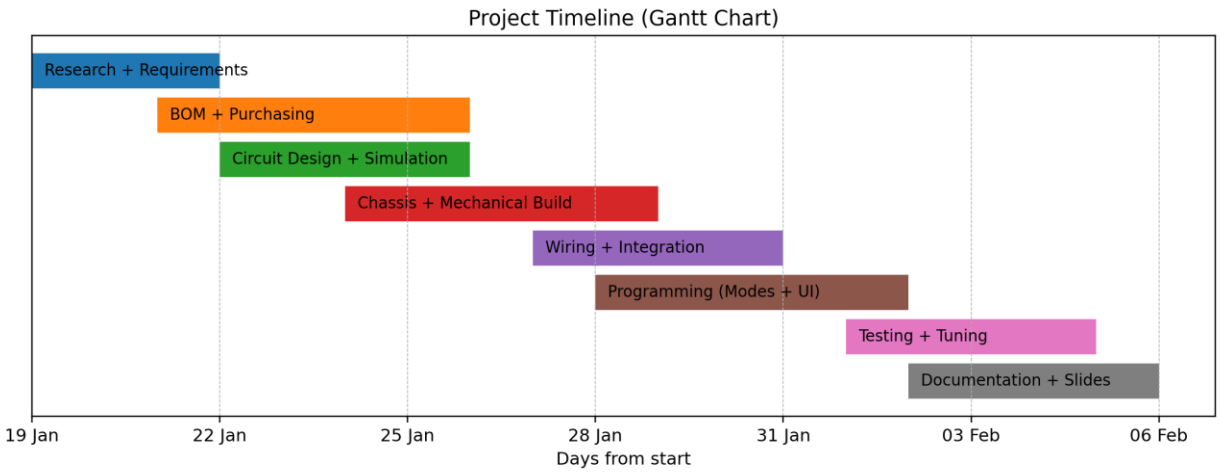
5. Flowchart

The following flowchart shows the main logic used in the robot program (mode selection, checks, and movement loop).


start
Power On Robot
Check Mode
Yes
Start Line Following
No
Random Mode
Obstacle Detect
User Hand Detect
If Yes , No
Yes
Open Lid
No
Wait 3s
Move
(Line / Random)
Close
Update Led Dispaly
end

6. Timeline (Gantt Chart)

The following timeline describes the planned work schedule from initial research to final testing and documentation.



7. Notes for Next Progress Meeting

- Complete chassis fabrication and mount the bin + lid mechanism.
- Finish wiring on a prototype board and validate stable 5V power using the buck converter.
- Implement and test dual-mode switching (LINE vs RANDOM) with safe obstacle avoidance.
- Calibrate line sensors and bin-full distance thresholds.
- Prepare short demo video and testing results (line-follow accuracy, obstacle avoid, lid response).